

LAB 3 — ARCHITECTURE SELECTION & JUSTIFICATION

Multi-Cloud Asset Cost Optimization Portal (MCACOP)

Architecture Selection: I chose Microservices Architecture for the Multi-Cloud Asset Cost Optimization Portal (MCACOP).

Architectural Choice: The selected architectural style is Microservices Architecture. MCACOP is divided into independently deployable services such as Billing Ingestion, Cost Analysis, Resource Optimization, Budget Monitoring and Alert Service, with the Portal UI acting as the user-facing component.

Reason 1 — Multi-cloud billing and analytics: MCACOP processes billing information from AWS, Azure and GCP and performs separate activities such as cost categorization, idle-compute detection, unattached-storage detection and savings estimation. Microservices allow these distinct responsibilities to remain modular and communicate through well-defined interfaces.

Reason 2 — Independent scalability: Different MCACOP workloads can require different amounts of processing. Billing ingestion and resource-optimization analysis may require more resources when large billing files or many cloud resources are processed, while dashboard and budget functions may have lighter workloads. Microservices allow individual services to be scaled independently.

Security Advantage: Microservices provide clear service boundaries where authentication and authorization controls can be applied to individual APIs. Administrative operations such as budget configuration can be restricted separately, and service-to-service communication can be secured using authenticated APIs.

Performance Benefit: Independent services can process workloads in parallel and can be scaled according to demand. For example, billing ingestion and resource-optimization analysis can operate independently instead of forcing all portal functions to share the same processing resources, improving responsiveness under heavier workloads.

Conclusion: Microservices Architecture fits MCACOP because its major functions are naturally separable and have different processing requirements. The architecture supports modular development, independent scaling, service-level security and parallel processing.